

1. Use simple CREATE Command:-

```
mysql> create database urmila;  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> use urmila;  
Database changed
```

```
mysql> create table student(Reg_no int, S_name char(20), Class varchar(10) ,Address varchar(30),phone_no int,DOB date);  
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> desc student;
```

Field	Type	Null	Key	Default	Extra
Reg_no	int	YES		NULL	
S_name	char(20)	YES		NULL	
Class	varchar(10)	YES		NULL	
Address	varchar(30)	YES		NULL	
phone_no	int	YES		NULL	
DOB	date	YES		NULL	

```
6 rows in set (0.08 sec)
```

INSERT Command

```
mysql> insert into student(Reg_no,S_name,Class,Address,phone_no,DOB) values(101,'Urmila','BCA','G.B nagar',99674534,'2006-08-27');
Query OK, 1 row affected (0.01 sec)
```

```
mysql> insert into student(Reg_no,S_name,Class,Address,phone_no,DOB) values(102,'Riya','BCA','Ghaziabad',98774534,'2005-07-25');
Query OK, 1 row affected (0.01 sec)
```

```
mysql> insert into student(Reg_no,S_name,Class,Address,phone_no,DOB) values(103,'Priya','BCA','Ghaziabad',98774674,'2006-06-23');
Query OK, 1 row affected (0.01 sec)
```

```
mysql> insert into student(Reg_no,S_name,Class,Address,phone_no,DOB) values(104,'Ashmita','BCA','Ghaziabad',98354674,'2007-05-22');
Query OK, 1 row affected (0.01 sec)
```

```
mysql> insert into student(Reg_no,S_name,Class,Address,phone_no,DOB) values(105,'Khushi','BCA','Noida',88354674,'2005-03-24');
Query OK, 1 row affected (0.01 sec)
```

```
mysql> select * from student;
```

Reg_no	S_name	Class	Address	phone_no	DOB
101	Urmila	BCA	G.B nagar	99674534	2006-08-27
102	Riya	BCA	Ghaziabad	98774534	2005-07-25
103	Priya	BCA	Ghaziabad	98774674	2006-06-23
104	Ashmita	BCA	Ghaziabad	98354674	2007-05-22
105	Khushi	BCA	Noida	88354674	2005-03-24

```
5 rows in set (0.00 sec)
```

Use CREATE Command with NOT NULL Key

```
mysql> create table customer(C_id int ,C_name char(20),Product_purchase char(30) not null,Product_qty int , C_address varchar(30));
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> desc customer;
```

Field	Type	Null	Key	Default	Extra
C_id	int	YES		NULL	
C_name	char(20)	YES		NULL	
Product_purchase	char(30)	NO		NULL	
Product_qty	int	YES		NULL	
C_address	varchar(30)	YES		NULL	

```
5 rows in set (0.00 sec)
```

INSERT

```
mysql> insert into customer(C_id,C_name,Product_purchase,Product_qty) values(201,'Shipra','Teddy Bear',2);
Query OK, 1 row affected (0.01 sec)
```

```
mysql> insert into customer(C_id,C_name,Product_purchase, C_address,Product_qty) values(202,'Anjali','Facewash','Ghaziabad',1);
Query OK, 1 row affected (0.01 sec)
```

```
mysql> insert into customer(C_id,C_name,Product_purchase, C_address) values(203,'Komal','water bottle','Ghaziabad');
Query OK, 1 row affected (0.01 sec)
```

```
mysql> insert into customer(C_id,C_name,Product_purchase, C_address,Product_qty) values(204,'Gunjan','Book','G.B nagar',4);
Query OK, 1 row affected (0.01 sec)
```

```
mysql> insert into customer(C_id,C_name,Product_purchase,Product_qty) values(205,'Shreya','Calculator',4);
Query OK, 1 row affected (0.01 sec)
```

Select command

```
mysql> select * from customer;
```

C_id	C_name	Product_purchase	Product_qty	C_address
201	Shipra	Teddy Bear	2	NULL
202	Anjali	Facewash	1	Ghaziabad
203	Komal	water bottle	NULL	Ghaziabad
204	Gunjan	Book	4	G.B nagar
205	Shreya	Calculator	4	NULL

```
5 rows in set (0.00 sec)
```

Use CREATE Command with UNIQUE Key

```
mysql> create table orders (Order_no int unique ,C_Name char(20),dopurchase date,D_address varchar(30),order_qty int);  
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> desc orders;
```

Field	Type	Null	Key	Default	Extra
Order_no	int	YES	UNI	NULL	
C_Name	char(20)	YES		NULL	
dopurchase	date	YES		NULL	
D_address	varchar(30)	YES		NULL	
order_qty	int	YES		NULL	

```
5 rows in set (0.00 sec)
```

```
mysql> insert into orders (order_no,c_name,dopurchase,order_qty) values(303,'Gita','2026-02-14',3);
```

```
Query OK, 1 row affected (0.01 sec)
```

```
mysql> insert into orders (order_no,c_name,dopurchase,D_address) values(304,'komal','2026-07-14','G.B nagar');
```

```
Query OK, 1 row affected (0.01 sec)
```

```
mysql> insert into orders (order_no,c_name,dopurchase,D_address) values(305,'Arti','2026-04-17','G.B nagar');
```

```
Query OK, 1 row affected (0.01 sec)
```

```
mysql> select * from orders;
```

Order_no	C_Name	dopurchase	D_address	order_qty
301	Urmila	2026-07-28	Ghaziabad	2
302	Gunjan	2026-08-29	Ghaziabad	NULL
303	Gita	2026-02-14	NULL	3
304	komal	2026-07-14	G.B nagar	NULL
305	Arti	2026-04-17	G.B nagar	NULL

```
5 rows in set (0.00 sec)
```

Use CREATE Command with Primary key:-

```
mysql> create table employee(E_id int primary key,E_name char(20),DOB date,DOJ date,E_salary int,Designation char(30));
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> desc employee;
```

Field	Type	Null	Key	Default	Extra
E_id	int	NO	PRI	NULL	
E_name	char(20)	YES		NULL	
DOB	date	YES		NULL	
DOJ	date	YES		NULL	
E_salary	int	YES		NULL	
Designation	char(30)	YES		NULL	

```
6 rows in set (0.00 sec)
```

```
mysql> insert into employee(E_id,E_name,DOB,DOJ,E_salary,Designation) values(401,'puja','2006-08-27','2026-07-24',30000,H.R);
ERROR 1054 (42S22): Unknown column 'H.R' in 'field list'
```

```
mysql> insert into employee(E_id,E_name,DOB,DOJ,E_salary,Designation) values(401,'puja','2006-08-27','2026-07-24',30000,worker);
ERROR 1054 (42S22): Unknown column 'worker' in 'field list'
```

```
mysql> insert into employee(E_id,E_name,DOB,DOJ,E_salary,Designation) values(401,'puja','2006-08-27','2026-07-24',30000,'worker');
Query OK, 1 row affected (0.01 sec)
```

```
mysql> insert into employee(E_id,E_name,DOB,DOJ,E_salary,Designation) values(402,'pinky','2008-07-26','2026-08-29',35000,'H.R');
Query OK, 1 row affected (0.01 sec)
```

```
mysql> insert into employee(E_id,E_name,DOB,DOJ,E_salary) values(403,'Rachna','2005-07-24','2025-08-23',25000);
Query OK, 1 row affected (0.01 sec)
```

```
mysql> select * from employee;
```

E_id	E_name	DOB	DOJ	E_salary	Designation
401	puja	2006-08-27	2026-07-24	30000	worker
402	pinky	2008-07-26	2026-08-29	35000	H.R
403	Rachna	2005-07-24	2025-08-23	25000	NULL
404	Tina	2007-06-20	2025-04-28	40000	NULL
405	nimrit	2007-07-24	2025-03-20	35000	NULL

```
5 rows in set (0.00 sec)
```

Use CREATE Command with check constraint key:-

```
mysql> create table persons(P_id int primary key,Name char(20),P_address varchar(30),Gender char(10),age int check(age>=18));
Query OK, 0 rows affected (0.04 sec)

mysql> desc persons;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| P_id  | int  | NO   | PRI | NULL    |       |
| Name  | char(20) | YES |     | NULL    |       |
| P_address | varchar(30) | YES |     | NULL    |       |
| Gender | char(10) | YES |     | NULL    |       |
| age   | int  | YES |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql> insert into persons(P_id,Name,P_address,Gender,age) values(501,'Monika','Ghaziabad','F',19);
Query OK, 1 row affected (0.01 sec)

mysql> insert into persons(P_id,Name,P_address,Gender,age) values(502,'Rita','G.B nagar','F',20);
Query OK, 1 row affected (0.01 sec)

mysql> insert into persons(P_id,Name,P_address,Gender,age) values(503,'Nandini','G.B nagar','F',21);
Query OK, 1 row affected (0.01 sec)

mysql> insert into persons(P_id,Name,P_address,Gender,age) values(504,'Pinky','Noida','F',18);
Query OK, 1 row affected (0.01 sec)

mysql> insert into persons(P_id,Name,P_address,Gender,age) values(505,'Sweety','Noida','F',22);
Query OK, 1 row affected (0.01 sec)

mysql> select * from persons;
+-----+-----+-----+-----+-----+
| P_id | Name  | P_address | Gender | age |
+-----+-----+-----+-----+-----+
| 501  | Monika | Ghaziabad | F      | 19  |
| 502  | Rita  | G.B nagar | F      | 20  |
| 503  | Nandini | G.B nagar | F      | 21  |
| 504  | Pinky  | Noida     | F      | 18  |
| 505  | Sweety | Noida     | F      | 22  |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

Use CREATE Command with Default key:-

```
mysql> create table student46(s_id int primary key,name char(20),DOB date,city char(30) default 'Ghaziabad',Gender char(10));
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> desc student46;
```

Field	Type	Null	Key	Default	Extra
s_id	int	NO	PRI	NULL	
name	char(20)	YES		NULL	
DOB	date	YES		NULL	
city	char(30)	YES		Ghaziabad	
Gender	char(10)	YES		NULL	

```
5 rows in set (0.00 sec)
```

```
mysql> insert into student46(s_id,name,DOB,Gender) values(601,'Lakshmi','2006-03-12','F');
Query OK, 1 row affected (0.01 sec)
```

```
mysql> insert into student46(s_id,name,DOB,Gender) values(602,'Neha','2005-04-24','F');
Query OK, 1 row affected (0.01 sec)
```

```
mysql> insert into student46(s_id,name,DOB,Gender) values(603,'Preeti','2005-06-22','F');
Query OK, 1 row affected (0.01 sec)
```

```
mysql> insert into student46(s_id,name,DOB,Gender) values(604,'Khushboo','2006-06-27','F');
Query OK, 1 row affected (0.01 sec)
```

```
mysql> insert into student46(s_id,name,DOB,Gender) values(605,'Khushi','2004-06-25','F');
Query OK, 1 row affected (0.01 sec)
```

```
mysql> select * from student46;
```

s_id	name	DOB	city	Gender
601	Lakshmi	2006-03-12	Ghaziabad	F
602	Neha	2005-04-24	Ghaziabad	F
603	Preeti	2005-06-22	Ghaziabad	F
604	Khushboo	2006-06-27	Ghaziabad	F
605	Khushi	2004-06-25	Ghaziabad	F

```
5 rows in set (0.00 sec)
```

1. Use UPDATE Command:-

```
mysql> select * from student;
```

Reg_no	S_name	Class	Address	phone_no	DOB
101	Urmila	BCA	G.B nagar	99674534	2006-08-27
102	Riya	BCA	Ghaziabad	98774534	2005-07-25
103	Priya	BCA	Ghaziabad	98774674	2006-06-23
104	Ashmita	BCA	Ghaziabad	98354674	2007-05-22
105	Khushi	BCA	Noida	88354674	2005-03-24

```
5 rows in set (0.00 sec)
```

```
mysql> update student set Address = 'GHAZIABAD';
```

```
Query OK, 5 rows affected (0.01 sec)
```

```
Rows matched: 5  Changed: 5  Warnings: 0
```

```
mysql> select * from student;
```

Reg_no	S_name	Class	Address	phone_no	DOB
101	Urmila	BCA	GHAZIABAD	99674534	2006-08-27
102	Riya	BCA	GHAZIABAD	98774534	2005-07-25
103	Priya	BCA	GHAZIABAD	98774674	2006-06-23
104	Ashmita	BCA	GHAZIABAD	98354674	2007-05-22
105	Khushi	BCA	GHAZIABAD	88354674	2005-03-24

```
5 rows in set (0.00 sec)
```

2. Use UPDATE Command with where:-

```
mysql> select * from customer;
```

C_id	C_name	Product_purchase	Product_qty	C_address
201	Shipra	Teddy Bear	2	NULL
202	Anjali	Facewash	1	Ghaziabad
203	Komal	water bottle	NULL	Ghaziabad
204	Gunjan	Book	4	G.B nagar
205	Shreya	Calculator	4	NULL

```
5 rows in set (0.00 sec)
```

```
mysql> update customer set product_purchase = 'Notebook' where C_id = 204;  
Query OK, 1 row affected (0.01 sec)  
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> select * from customer;
```

C_id	C_name	Product_purchase	Product_qty	C_address
201	Shipra	Teddy Bear	2	NULL
202	Anjali	Pen	1	Ghaziabad
203	Komal	water bottle	NULL	Ghaziabad
204	Gunjan	Notebook	4	G.B nagar
205	Shreya	Calculator	4	NULL

```
5 rows in set (0.00 sec)
```

1. Use ALTER Command With ADD Column:-

```
mysql> alter table customer add email_id varchar(20);
```

```
Query OK, 0 rows affected (0.02 sec)
```

```
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> desc customer;
```

Field	Type	Null	Key	Default	Extra
C_id	int	YES		NULL	
C_name	char(20)	YES		NULL	
Product_purchase	char(30)	NO		NULL	
Product_qty	int	YES		NULL	
C_address	varchar(30)	YES		NULL	
email_id	varchar(20)	YES		NULL	

```
6 rows in set (0.00 sec)
```

```
mysql> update customer set email_id = 'shipra123@gmail.com' where C_id = 201;
```

```
Query OK, 1 row affected (0.01 sec)
```

```
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> select * from customer;
```

C_id	C_name	Product_purchase	Product_qty	C_address	email_id
201	Shipra	Teddy Bear	2	NULL	shipra123@gmail.com
202	Anjali	Pen	1	Ghaziabad	NULL
203	Komal	water bottle	NULL	Ghaziabad	NULL
204	Gunjan	Notebook	4	G.B nagar	NULL
205	Shreya	Calculator	4	NULL	NULL

```
5 rows in set (0.00 sec)
```

2. Use ALTER Command with DELETE Column:-

```
mysql> select * from customer;
```

C_id	C_name	Product_purchase	Product_qty	C_address	email_id
201	Shipra	Teddy Bear	2	NULL	shipra123@gmail.com
202	Anjali	Pen	1	Ghaziabad	NULL
203	Komal	water bottle	NULL	Ghaziabad	NULL
204	Gunjan	Notebook	4	G.B nagar	NULL
205	Shreya	Calculator	4	NULL	NULL

```
5 rows in set (0.00 sec)
```

```
mysql> alter table customer drop email_id;
```

```
Query OK, 0 rows affected (0.02 sec)
```

```
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> select * from customer;
```

C_id	C_name	Product_purchase	Product_qty	C_address
201	Shipra	Teddy Bear	2	NULL
202	Anjali	Pen	1	Ghaziabad
203	Komal	water bottle	NULL	Ghaziabad
204	Gunjan	Notebook	4	G.B nagar
205	Shreya	Calculator	4	NULL

```
5 rows in set (0.00 sec)
```

3. USE ALTER Command with RENAME Column:-

```
mysql> select * from customer;
```

C_id	C_name	Product_purchase	Product_qty	C_address
201	Shipra	Teddy Bear	2	NULL
202	Anjali	Pen	1	Ghaziabad
203	Komal	water bottle	NULL	Ghaziabad
204	Gunjan	Notebook	4	G.B nagar
205	Shreya	Calculator	4	NULL

```
5 rows in set (0.00 sec)
```

```
mysql> alter table customer rename column C_address to city;
```

```
Query OK, 0 rows affected (0.02 sec)
```

```
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> select * from customer;
```

C_id	C_name	Product_purchase	Product_qty	city
201	Shipra	Teddy Bear	2	NULL
202	Anjali	Pen	1	Ghaziabad
203	Komal	water bottle	NULL	Ghaziabad
204	Gunjan	Notebook	4	G.B nagar
205	Shreya	Calculator	4	NULL

```
5 rows in set (0.00 sec)
```

4. Use ALTER Command with MODIFY Column:-

```
mysql> desc customer;
```

Field	Type	Null	Key	Default	Extra
C_id	int	YES		NULL	
C_name	char(20)	YES		NULL	
Product_purchase	char(30)	NO		NULL	
Product_qty	int	YES		NULL	
city	varchar(30)	YES		NULL	

```
5 rows in set (0.00 sec)
```

```
mysql> alter table customer modify city char(20);
```

```
Query OK, 5 rows affected (0.04 sec)
```

```
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> select * from customer;
```

C_id	C_name	Product_purchase	Product_qty	city
201	Shipra	Teddy Bear	2	NULL
202	Anjali	Pen	1	Ghaziabad
203	Komal	water bottle	NULL	Ghaziabad
204	Gunjan	Notebook	4	G.B nagar
205	Shreya	Calculator	4	NULL

```
5 rows in set (0.00 sec)
```

```
mysql> desc customer;
```

Field	Type	Null	Key	Default	Extra
C_id	int	YES		NULL	
C_name	char(20)	YES		NULL	
Product_purchase	char(30)	NO		NULL	
Product_qty	int	YES		NULL	
city	char(20)	YES		NULL	

USE DELETE Command

```
mysql> select * from persons;
```

P_id	Name	P_address	Gender	age
501	Monika	Ghaziabad	F	19
502	Rita	G.B nagar	F	20
503	Nandini	G.B nagar	F	21
504	Pinky	Noida	F	18
505	Sweety	Noida	F	22

```
5 rows in set (0.00 sec)
```

```
mysql> delete from persons where P_id = 505;
```

```
Query OK, 1 row affected (0.02 sec)
```

```
mysql> select * from persons;
```

P_id	Name	P_address	Gender	age
501	Monika	Ghaziabad	F	19
502	Rita	G.B nagar	F	20
503	Nandini	G.B nagar	F	21
504	Pinky	Noida	F	18

```
4 rows in set (0.00 sec)
```

USE RENAME Commands

```
mysql> select * from persons;
```

P_id	Name	P_address	Gender	age
501	Monika	Ghaziabad	F	19
502	Rita	G.B nagar	F	20
503	Nandini	G.B nagar	F	21
504	Pinky	Noida	F	18

```
4 rows in set (0.00 sec)
```

```
mysql> rename table persons to client;
```

```
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> select * from client;
```

P_id	Name	P_address	Gender	age
501	Monika	Ghaziabad	F	19
502	Rita	G.B nagar	F	20
503	Nandini	G.B nagar	F	21
504	Pinky	Noida	F	18

```
4 rows in set (0.00 sec)
```

USE TRUNCATE Command:-

```
mysql> truncate table student;  
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> select * from student;  
Empty set (0.00 sec)
```

```
mysql> desc student;
```

Field	Type	Null	Key	Default	Extra
Reg_no	int	YES		NULL	
S_name	char(20)	YES		NULL	
Class	varchar(10)	YES		NULL	
Address	varchar(30)	YES		NULL	
phone_no	int	YES		NULL	
DOB	date	YES		NULL	

```
6 rows in set (0.00 sec)
```

```
mysql> |
```

Use DROP Command:-

```
mysql> desc student;
```

Field	Type	Null	Key	Default	Extra
Reg_no	int	YES		NULL	
S_name	char(20)	YES		NULL	
Class	varchar(10)	YES		NULL	
Address	varchar(30)	YES		NULL	
phone_no	int	YES		NULL	
DOB	date	YES		NULL	

```
6 rows in set (0.00 sec)
```

```
mysql> drop table student;  
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> desc student;  
ERROR 1146 (42S02): Table 'urmila.student' doesn't exist  
mysql> |
```